

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Use the de Broglie relation for momentum : $\lambda = h/(mv)$. The kinetic energy 1 eV gives the momentum by $p = \sqrt{2mE}$, and substituting in gives $\lambda = h/(\sqrt{2mE})$. Using $h = 6.625 \times 10^{-27}$ erg $\cdot$ s , $m = 9.11 \times 10^{-28}$ g , and $E = 1.602 \times 10^{-12}$ erg , we get $\lambda \approx 6.625 \times 10^{-11}$ m = 6.63×10^{-8} cm .

Correct answer: F. 5.27×10^{-9} m (equivalently 6.63×10^{-8} cm).

</think>
<answer>
F. 5.27×10^{-9} m
</answer>

